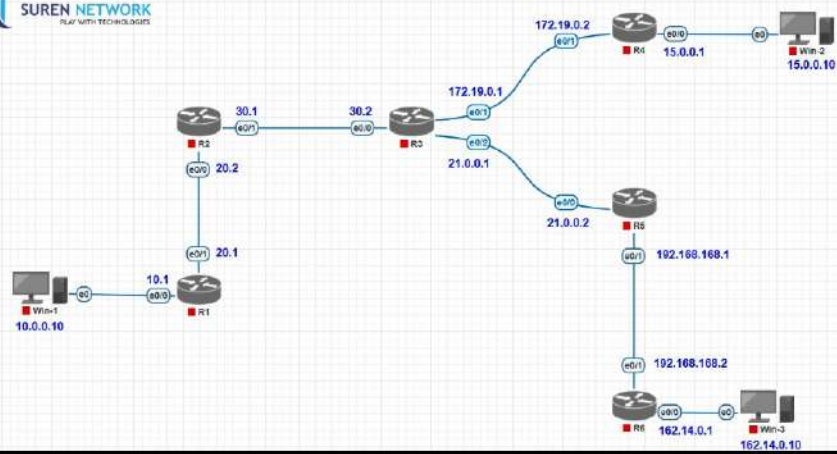
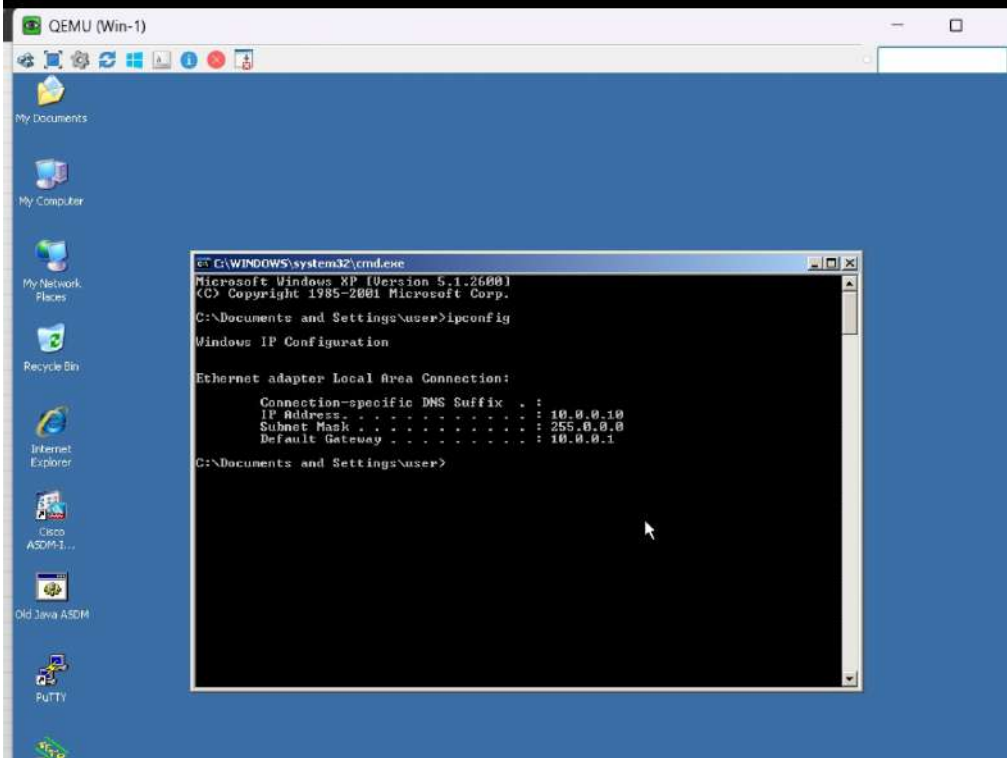






```
R1
S    172.19.0.0/16 [1/0] via 20.0.0.2
S    192.168.168.0/24 [1/0] via 20.0.0.2
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#
R1(config)#exit
R1#
*Aug 23 11:30:38.315: %SYS-5-CONFIG_I: Configured from console by console
R1#
R1#
R1#show ip int br

```

Interface	IP-Address	OK?	Method	Status	Prot
Loopback0					
Ethernet0/0	10.0.0.1	YES	manual	up	up
Ethernet0/1	20.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S       15.0.0.0/8 [1/0] via 20.0.0.2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S       21.0.0.0/8 [1/0] via 20.0.0.2
S       30.0.0.0/8 [1/0] via 20.0.0.2
S       162.14.0.0/16 [1/0] via 20.0.0.2
S       172.19.0.0/16 [1/0] via 20.0.0.2
S       192.168.168.0/24 [1/0] via 20.0.0.2
R1#

```

 R2—□×

```
R2#
R2#
R2#show ip int br
```

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	20.0.0.2	YES	manual	up	up
Ethernet0/1	30.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down



```

R2#
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     20.0.0.0/8 is directly connected, Ethernet0/0
L     20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/1
L     30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2
--More--

```

R3

```
R3#
R3#
R3#show ip int br
Interface      IP-Address      OK? Method Status      Prot
loco1
Ethernet0/0    30.0.0.2        YES manual up          up
Ethernet0/1    172.19.0.1      YES manual up          up
Ethernet0/2    21.0.0.1        YES manual up          up
Ethernet0/3    unassigned      YES TFTP  down        down
```

```

R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

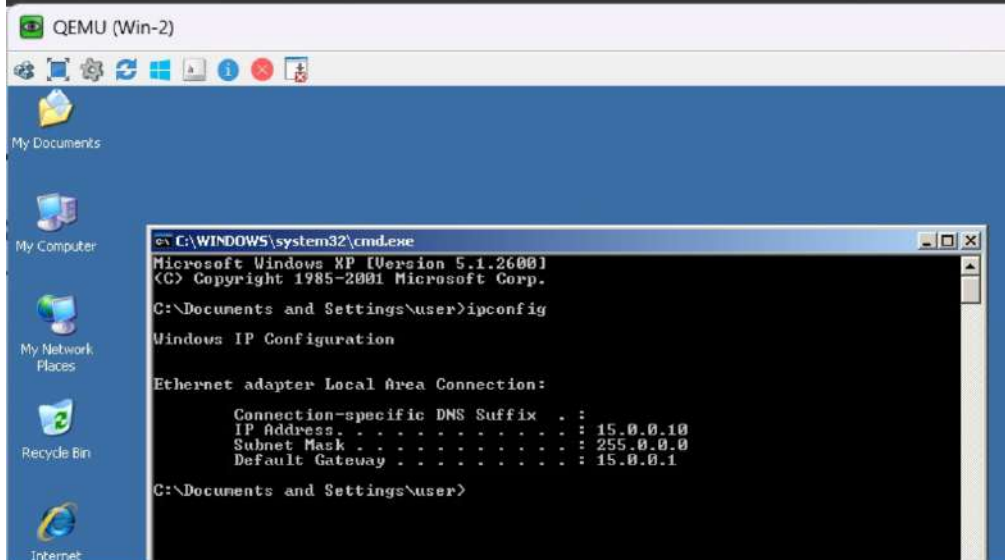
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/2
L    21.0.0.1/32 is directly connected, Ethernet0/2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    30.0.0.0/8 is directly connected, Ethernet0/0
L    30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
     172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
--More--

```

```
R4
R4#show ip int br
Interface              IP-Address      OK? Method Status      Protocol
Ethernet0/0            15.0.0.1        YES manual up          up
Ethernet0/1            172.19.0.2      YES manual up          up
Ethernet0/2            unassigned      YES TFTP  down        down
Ethernet0/3            unassigned      YES TFTP  down        down
R4#
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
--More--
```



```

R5#
R5#show ip int br
Interface                IP-Address      OK? Method Status        Protocol
Ethernet0/0              21.0.0.2        YES manual up             up
Ethernet0/1              192.168.168.1   YES manual up             up
Ethernet0/2              unassigned      YES TFTP  down           down
Ethernet0/3              unassigned      YES TFTP  down           down
R5#
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

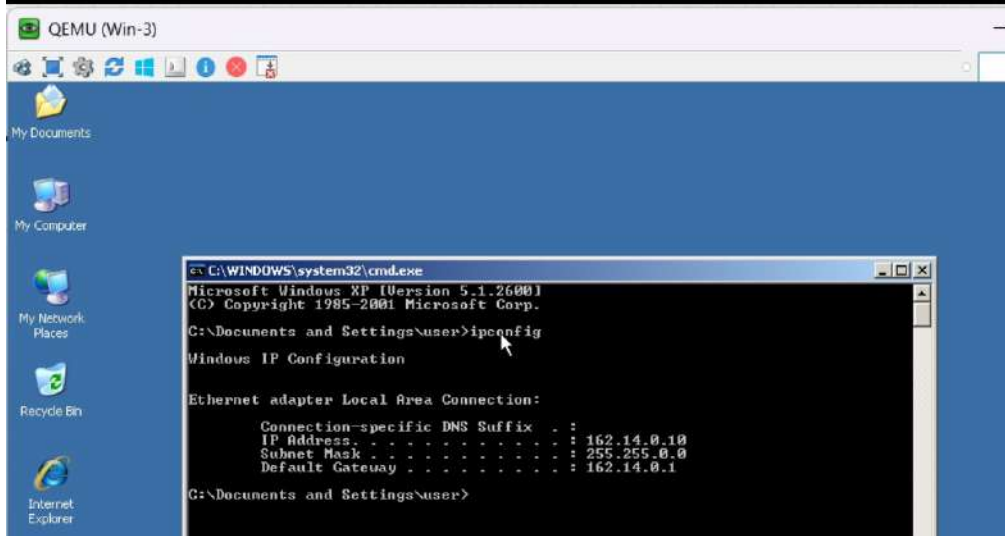
S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
S    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/0
L    21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
S    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.1/32 is directly connected, Ethernet0/1
R5#
R5#

```

```
R6#
R6#show ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              162.14.0.1      YES manual up          up
Ethernet0/1              192.168.168.2   YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R6#
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PBR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
C    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
L    162.14.0.0/16 is directly connected, Ethernet0/0
L    162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
C    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.2/32 is directly connected, Ethernet0/1
R6#
R6#
```



```
C:\WINDOWS\system32\cmd.exe

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 10.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 10.0.0.1

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=4ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=4ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>_
```

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 15.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 15.0.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```

```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 162.14.0.10
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 162.14.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```

